AxSure™ Sustainable Destruction of Hazardous Wastes

Solution name

AxSure™ Sustainable Destruction of Hazardous Wastes

Category

2026 Nominations

☐ Please tick if you have previously been nominated to The Earthshot Prize or already shared your information with us this year.

If you have previously been nominated to The Earthshot Prize, we aim to save you time. Please tick the box above and you may skip any answers below if you feel your last nomination already provides answers that are up to date.

What is the specific environmental challenge that your work addresses?

126 / 150 words

AxNano is confronting the global crisis of persistent chemical and microplastic pollution, which threatens ecosystems and human health. Industrial waste, incineration, and landfilling release contaminants like PFAS, found in 97% of Americans' blood and exceeding safety limits in 69% of Canadian groundwater, and microplastics, with up to 40 million tons entering the environment annually. These pollutants bioaccumulate, disrupt wildlife, and pose severe risks, especially to vulnerable communities. Current waste management perpetuates contamination, but AxNano's AxSure™ technology offers a transformative solution, using supercritical water oxidation to destroy hazardous wastes at their source, achieving 99.999% PFAS removal from wastestreams that would otherwise lead to environmental release, ecosystem exposure. By replacing outdated practices, AxNano aims to restore ecosystems, protect biodiversity, and secure a sustainable future for generations to come.

What is the urgent challenge that you are working on that directly impacts one or more Earthshots? Please describe the scale of the challenge including numbers where possible.

E.g. Bogotá has long battled air pollution caused by diesel public buses, freight trucks, and dust from unpaved roads, especially in its most vulnerable neighbourhoods.

Between 1998 and 2005, Bogotá's air quality reached critical levels. Concentrations of particulate matter exceeded seven times the limit currently set out by the World Health Organisation. This had a severe impact on public health, particularly amongst children, the elderly, and low-income communities.

Urban air pollution is a health issue for every city on Earth. For more than 40% of global cities, the challenge is significant, with pollutant levels many times higher than is safe.

Max words 150

Please describe your solution and how it addresses the challenge in non expert language.

257 / 250 words

AxNano's solution, AxSure™, uses a proven process called supercritical water oxidation, which has been safely used for over 50 years to destroy hazardous chemicals during demilitarization activities, such as disposing of military waste. We are repurposing this technology to address today's environmental crisis, focusing on eliminating dangerous chemicals and microplastics before they can pollute our water, soil, or air. This process is not only trusted for large-scale operations, but it was also funded by the Gates Foundation as part of an alternative toilet project, proving its effectiveness on both large and small scales.

AxSure™ works by using elevated temperature and pressure, much higher than normal water treatment conditions, to unlock unique chemistries. This allows us to harness the special properties of supercritical water and reactive oxygen species, breaking down contaminants faster and more completely than any other practice. Instead of burning or burying waste, which often releases even more harmful substances, AxSure™ transforms pollutants, mineralizing them into harmless materials. This means we can remove almost all of the "forever chemicals," like PFAS, and microplastics from waste streams that would otherwise end up in the environment and threaten wildlife and human health.

To accelerate our impact and respond to the urgency of the problem, AxNano has formed a strategic partnership with a contractor who has decades of experience operating these systems for demilitarization. This partnership allows us to deliver solutions quickly and efficiently, leveraging their expertise to ensure safe and reliable operations. AxSure™ helps industries stop pollution at its source, protects communities and ecosystems, and supports a cleaner, healthier planet.

Please support your description with practical insights on how your solutions works and include why your solution and type of solution is most appropriate to address this challenge.

e.g. Tenure Facility is a global funding and legal mechanism dedicated to guaranteeing land and forest rights for Indigenous and, Afrodescendant Peoples, as well as local communities.

By combining legal expertise with direct funding, Tenure Facility provides communities with the tools they need to secure their land rights, safeguarding ancestral homelands and ensuring long-term stewardship of forests and ecosystems.

Max words 250

What is your 2030 vision?

165 / 150 words

By 2030, we envision a world where there is a solution for sustainable management of chemical and microplastic wastes. Entities will have a realistic option to choose as an alternative to landfilling and incineration, making advanced waste destruction technologies like AxSure™ accessible and practical. Our goal is to have four AxSure™ units deployed in the US, strategically located near military and private industries known to generate PFAS waste so that we can have the highest impact. We anticipate having up to two units in Canada, likely supporting the oil and gas industry. Lastly, we anticipate having 5 units across the EU, with faster uptake driven by stricter environmental regulations. We intend for AxSure™ to become the standard practice for waste destruction, enabling industries, communities, and governments to eliminate “forever chemicals” and microplastics at their source using proven processes such as supercritical water oxidation. Through strategic partnerships and innovation, AxNano aims to restore clean water, healthy soil, and fresh air, supporting resilient ecosystems and communities worldwide.

What does success look like when your solution reaches its full potential? Why is your work potentially transformative for the challenge you are addressing?

MAX 150 WORDS

What is the one unlock that would accelerate progress in your sector?

218 / 250 words

One transformative unlock for accelerating progress in chemical and microplastic waste destruction is dedicated funding through strategic partnerships with industry-aligned leaders. If innovators like AxNano join forces with major waste management companies such as Veolia or other international players, we can rapidly build, deploy, and demonstrate advanced waste destruction units like AxSure™ at scale. These partnerships would not only provide essential financial resources, but also bring together deep operational expertise, regulatory insight, logistics capabilities, and established customer trust.

By leveraging shared knowledge, industry partners can help navigate complex permitting processes, optimize supply chains, and ensure that new technologies meet real-world needs. Their experience with regulations and logistics would streamline deployment, while their reputation and relationships with customers would foster confidence in adopting new solutions. Demonstration projects, supported by these partnerships, would validate performance, generate real-world data, and showcase the effectiveness of supercritical water oxidation for eliminating hazardous pollutants.

This collaborative approach would overcome barriers to adoption, accelerate learning, and create momentum for sector-wide change. With industry leaders championing proven solutions, the sector could transition away from outdated practices, expedite the elimination of “forever chemicals” and microplastics, and set a new standard for sustainable waste management. Ultimately, strategic partnerships would drive innovation, scale impact, and deliver tangible benefits for communities and the environment, making advanced waste destruction a practical reality worldwide.

Please outline what levers could be utilised to accelerate scale in your sector. These could include; behaviour and culture shifts, technological innovation, market forces, changes in policy and regulation

250 WORDS MAX

What environmental benefit has your solution delivered to date?

- [x] GHG Emissions saved, captured or avoided e.g. tonnes of CO2 emissions avoided
- [x] Waste saved, reduced, recycled or avoided e.g. tonnes of plastic waste recycled
- [x] Water saved, reduced, recycled or avoided e.g. litres of water recycled
- [x] Reduction in air pollution e.g. % change in PM 2.5/10
- [ ] Area of land, ocean or water protected, restored or rewilded e.g. hectares of forest restored
- [ ] Biodiversity increases in land, sea or rivers e.g. increase in number of fish species

☐ Additional Described Field

What environmental benefit has your solution delivered to date? (optional)

	Metric	Impact
1	GHG Emissions saved, captured or avoided e.g. tonnes of CO2 emissions avoided	72 tonnes CO2e to date
2	Waste saved, reduced, recycled or avoided e.g. tonnes of plastic waste recycled	74 tonnes of industrial was
3	Water saved, reduced, recycled or avoided e.g. litres of water recycled	72,600 L of water recycled
4	Reduction in air pollution e.g. % change in PM 2.5/10	1852 gram PM2.5 and PM1
5	Area of land, ocean or water protected, restored or rewilded e.g. hectares of forest restored	
6	Biodiversity increases in land, sea or rivers e.g. increase in number of fish species	
7	Other	

Add row

To add an additional metric, click 'Add row'
Please add numbers where possible, if no numbers are available please add a comment

What environmental benefit could your solutions deliver in 5 years? (optional)

	Metric	Impact
1	GHG Emissions saved, captured or avoided e.g. tonnes of CO2 emissions avoided	23,550 tonnes CO2e avoid
2	Waste saved, reduced, recycled or avoided e.g. tonnes of plastic waste recycled	241,200 tonnes
3	Water saved, reduced, recycled or avoided e.g. litres of water recycled	236,376,000 L of water rec
4	Reduction in air pollution e.g. % change in PM 2.5/10	446,900 kg PM 2.5 and PM
5	Area of land, ocean or water protected, restored or rewilded e.g. hectares of forest restored	
6	Biodiversity increases in land, sea or rivers e.g. increase in number of fish species	
7	Other	

Add row

To add an additional metric, click 'Add row'
Please add numbers where possible, if no numbers are available please add a comment

What social benefit has your solution delivered to date?

- [x] Jobs created e.g. people employed to work on the solution
- [ ] Shared economic prosperity e.g. farmers whose income increases from using the solution
- [x] Lives improved e.g. other beneficiaries or end users who benefit
- [ ] Financial inclusion in appropriate financial services for underserved populations
- [ ] Policy influenced e.g. new government policies implemented, changed or adopted
- [ ] Additional Described Field

What social benefit has your solution delivered to date ? (optional)

	Metric	Impact
1	Jobs created e.g. people employed to work on the solution	10
2	Shared economic prosperity e.g. farmers whose income increases from using the solution	
3	Lives improved e.g. other beneficiaries or end users who benefit	
4	Financial inclusion in appropriate financial services for underserved populations	
5	Policy influenced e.g. new government policies implemented, changed or adopted	
6	Other	

Add row

To add an additional metric, click 'Add row'
Please add numbers where possible, if no numbers are available please add a comment

What social benefit could your solution deliver in 5 years? (optional)

	Metric	Impact
1	Jobs created e.g. people employed to work on the solution	
2	Shared economic prosperity e.g. farmers whose income increases from using the solution	
3	Lives improved e.g. other beneficiaries or end users who benefit	
4	Financial inclusion in appropriate financial services for underserved populations	
5	Policy influenced e.g. new government policies implemented, changed or adopted	
6	Other	

Add row

To add an additional metric, click 'Add row'
Please add numbers where possible, if no numbers are available please add a comment

How has your work to date been supported?

- [ ] Self-Funded
- [ ] Philanthropic

- [x] Angel Investors
- [x] Equity
- [x] Debt
- [ ] Project Financing
- [ ] In Kind Support
- [ ] Partnerships
- [ ] Grants
- [x] Other

Please explain how much traction your work has received through funding, partnerships, commercial relationships and publicity to date.

170 / 200 words

AxNano has achieved significant traction through funding, partnerships, and endorsements. For PFAS destruction technology, AxNano raised $2.4M private equity and, in October 2025, confirmed $2M from Leonid Capital under the Air Force TACFI program, leveraging a $2M capital match to attract $3M additional investment. A $2M TACFI award was announced in August 2025, alongside participation in the Army xTech Innovator's Corner at AUSA and pursuit of the $14M Army CATALYST program. AxNano is planning a $7M equity round in 2026 to scale AxSure™ SCWO systems.

Current partners and major relationships include General Atomics (commercialization agreement signed August 2024), Geosyntec Consultants (technical subcontract), and academic collaborators at Duke University, NC A&T, and UNC Chapel Hill. AxNano is in discussions with a large public entity in the waste management space regarding a strategic investment and 12-month PFAS destruction pilot.

Awards and endorsements: Multiple SBIR Phase I/II awards, EPA contracts, AFWERX TACFI, and One NC Matching Grant.

Include details of any funding, awards, collaborations, partnerships, significant media features, or other endorsements that you've received, and dates when they took place. Please list your current partners and relationships with major customers, grant funders, or suppliers.

MAX 200 WORDS

Which of the following options best describes the stage the solution is at?

Early deployment: The solution (product, service, policy, programme etc) has recently started to be deployed in a real world setting and

Please select one

Solution Type

Technology / Data-based solution

Please select one

If you selected other above, please describe

0 / 20 words

Organisation Type

For-profit organisation (privately held)

Please select one

What is your financial turnover?

$1,000,001 - $10,000,000 USD

Organisation Headquarters Location

United States of America (the)

Country or Territory where the organisation is registered

Geographical region of primary operating activities

North America

Country or territory of primary operating activities

United States of America

Please select any additional operating countries or territories (optional)

Search list

- [ ] Select all
- [ ] Turkmenistan
- [ ] Turks and Caicos Islands (the)
- [ ] Tuvalu
- [ ] Uganda
- [ ] Ukraine

Please confirm your number of employees

1-10 employees

Is your solution woman led?

50% of Management Team/Co-Founders

Do members of the leadership team belong to an indigenous community, is this solution indigenous-led?

No

Indigenous-led or First Nations solutions place Indigenous cultural values and knowledge at the centre of activity implementation. These initiatives include efforts to protect Indigenous lands or are led by individuals who identify as part of an Indigenous community.

Secondary Contact First Name

Doug

We encourage you to provide a secondary contact for us to reach out to, should the primary contact be unavailable.

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Facebook (optional)

https://

X (optional)

https://

LinkedIn (optional)

https://

YouTube (optional)

https://

TikTok (optional)

https://

Other Platforms (optional)

https://

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